

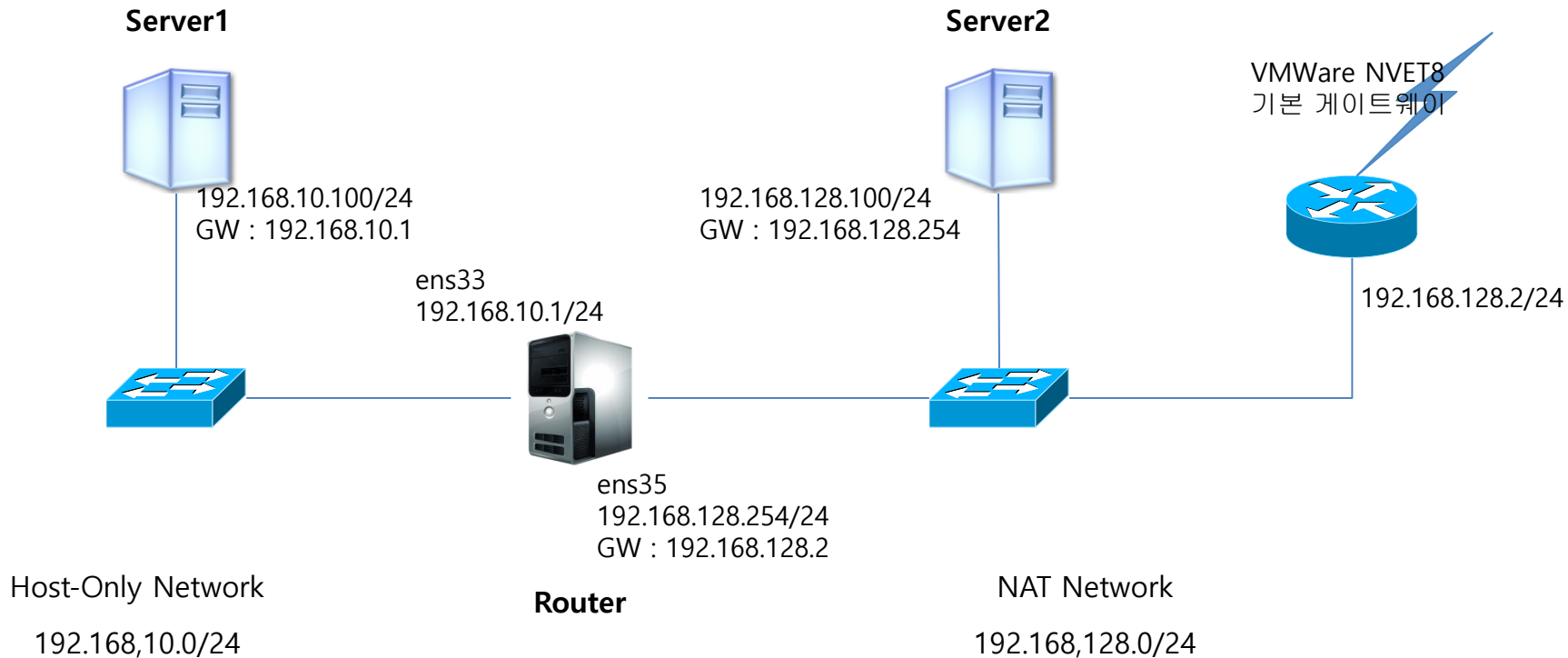
네트워크 실무

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1. LINUX 라우터

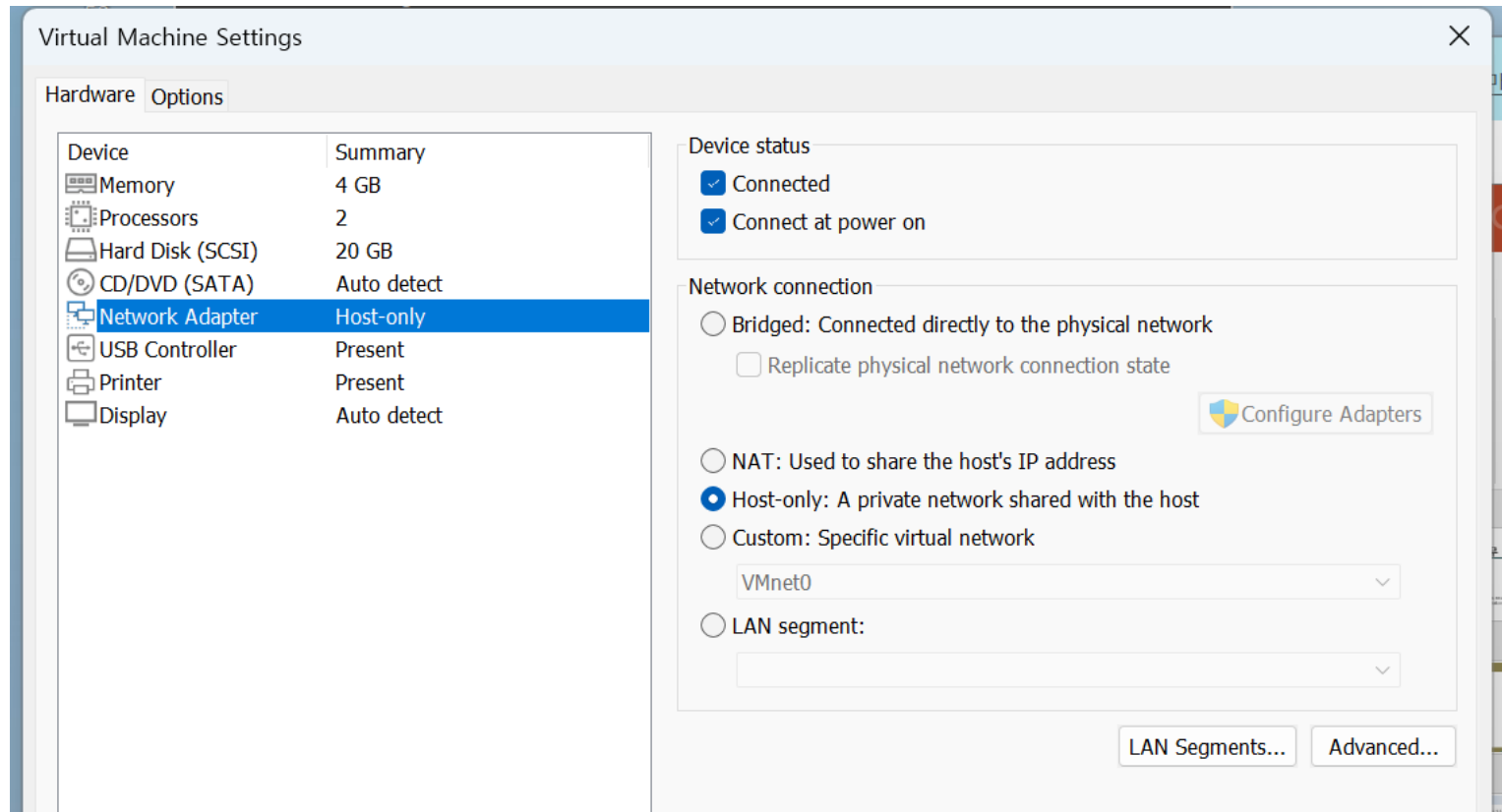
구성도

- 우분투 리눅스로 라우터를 만들어 서로 다른 네트워크간 통신을 할 수 있도록 설정하도록 합니다.



Server1 설정

■ Network Adapter : Host-Only



Server1 설정

- Ubuntu Network 설정
 - IPv4 Method : Manual
 - IP : 192.168.10.100
 - Netmask : 24
 - Gateway : 192.168.10.1

Cancel **Wired** Apply

Details Identity **IPv4** IPv6 Security

IPv4 Method

☐ Automatic (DHCP) ☐ Link-Local Only

☒ **Manual** ☐ Disable

☐ Shared to other computers

Addresses

Address	Netmask	Gateway	
192.168.10.100	255.255.255.0	192.168.10.1	

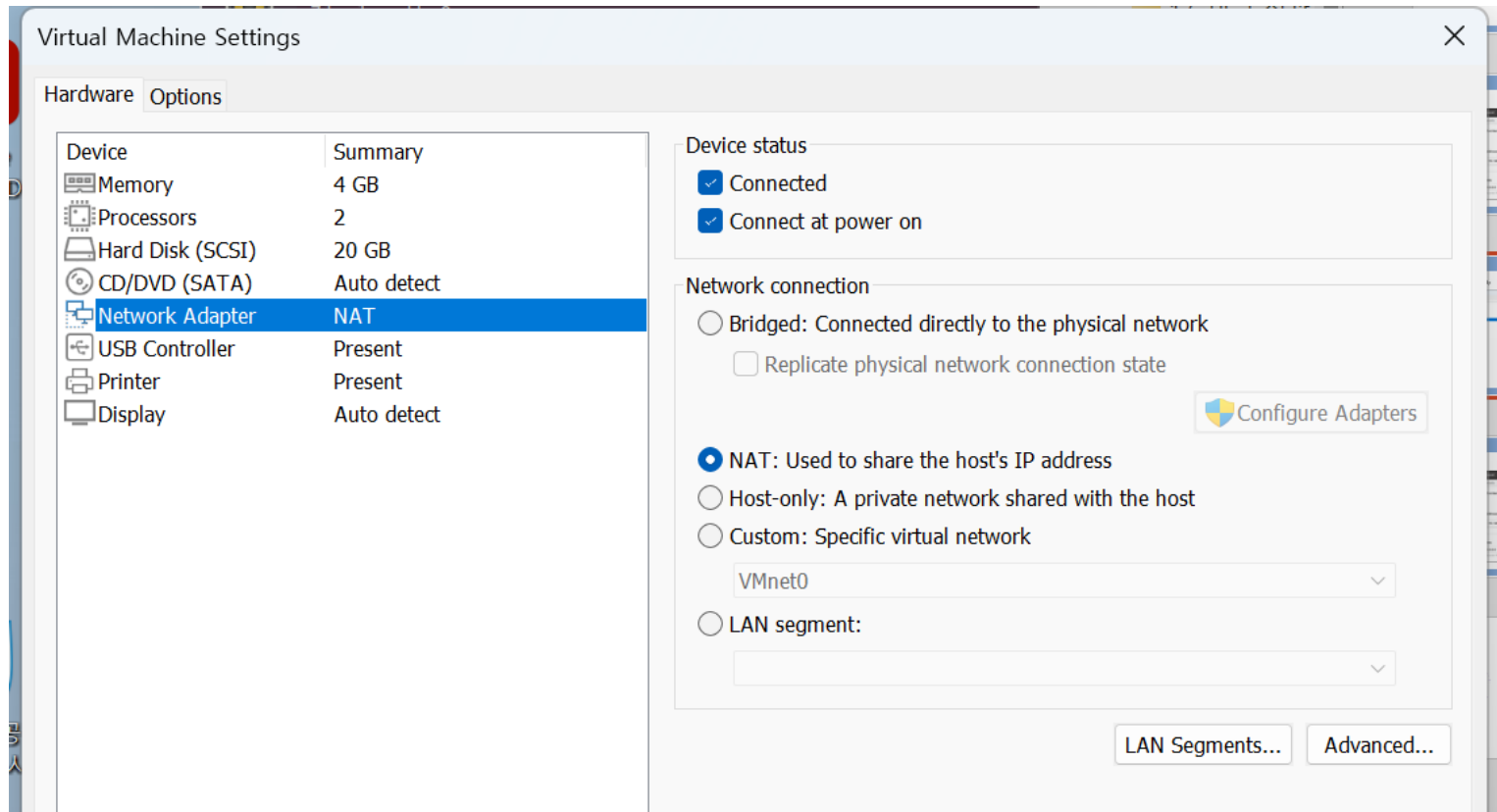
DNS Automatic ☒

8.8.8.8

Separate IP addresses with commas

Server2 설정

■ Network Adapter : Host-Only



Server2 설정

- Ubuntu Network 설정
 - IPv4 Method : Manual
 - IP : 192.168.128.100
 - Netmask : 24
 - Gateway : 192.168.128.1
 - DNS : 8.8.8.8

Cancel **Wired** Apply

Details Identity **IPv4** IPv6 Security

IPv4 Method

☐ Automatic (DHCP) ☐ Link-Local Only

☒ **Manual** ☐ Disable

☐ Shared to other computers

Addresses

Address	Netmask	Gateway	
192.168.128.100	255.255.255.0	192.168.128.254	

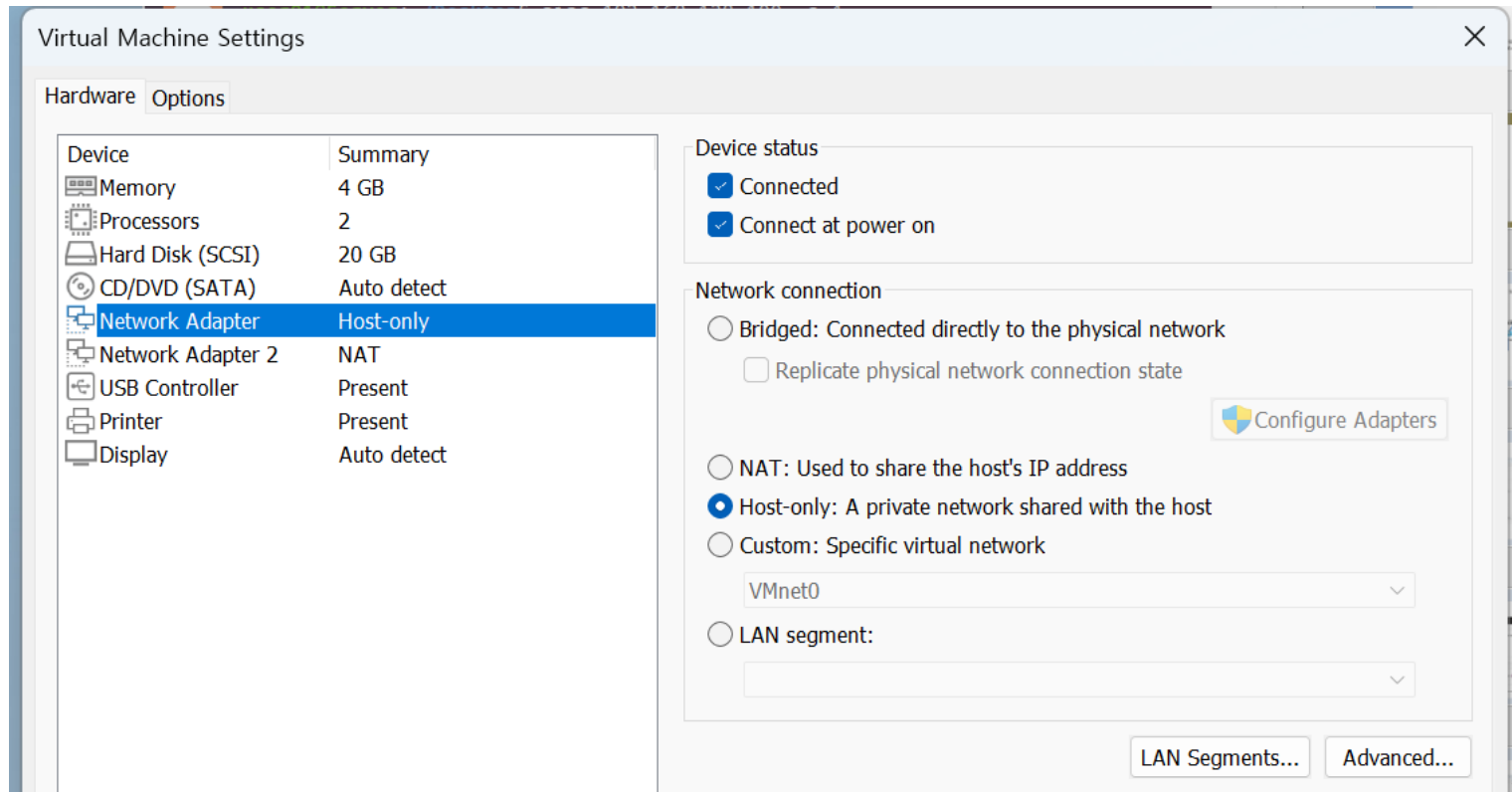
DNS Automatic ☒

8.8.8.8

Separate IP addresses with commas

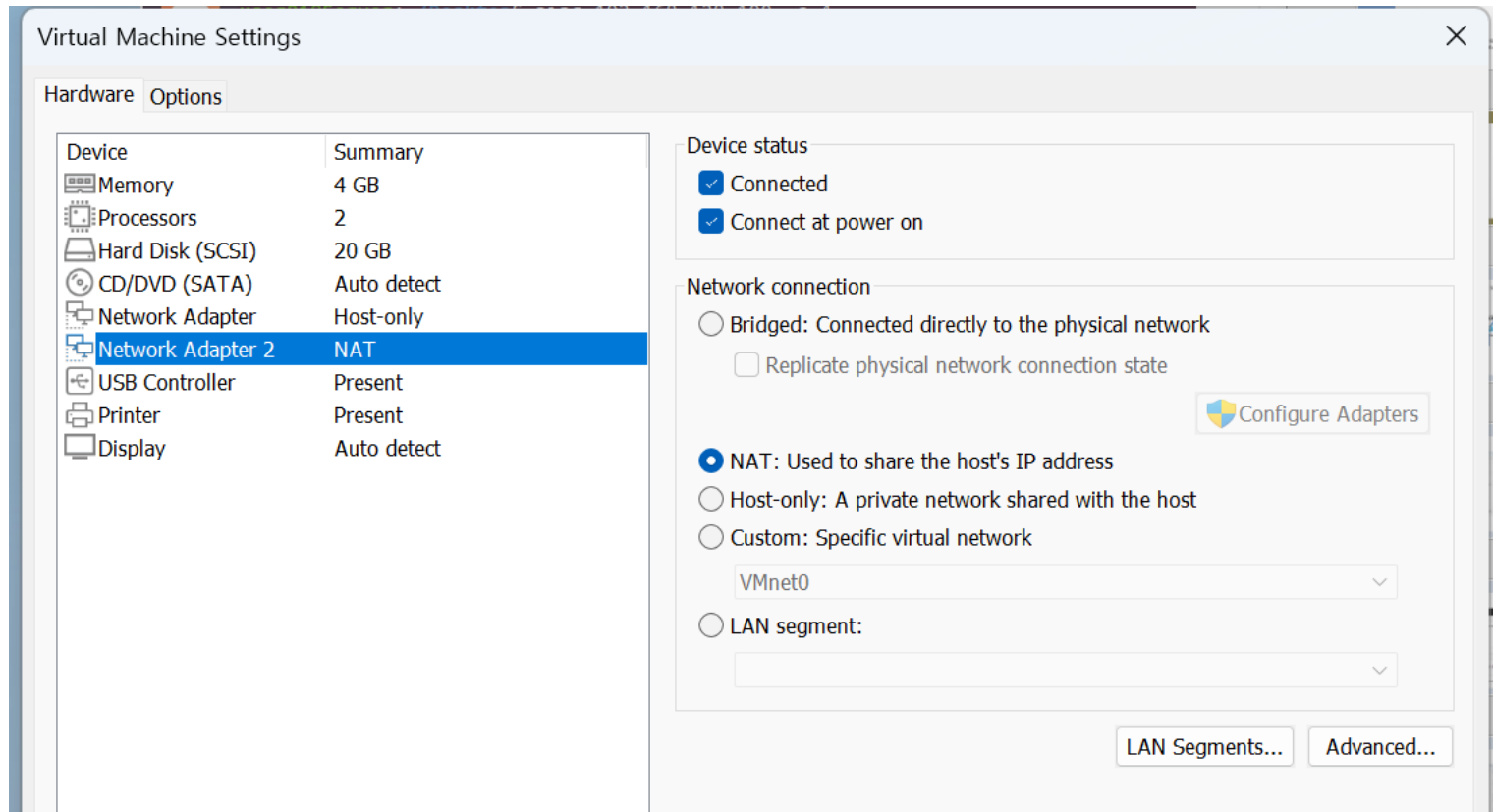
Router 설정

■ Network Adapter : Host-Only



Router 설정

■ Network Adapter 2 : NAT



Router 설정

- Ubuntu Network 설정
 - Interface : ens33
 - IPv4 Method : Manual
 - IP : 192.168.10.1
 - Netmask : 24
 - DNS : 8.8.8.8

Cancel **Wired** Apply

Details Identity **IPv4** IPv6 Security

IPv4 Method

☐ Automatic (DHCP) ☐ Link-Local Only

☒ **Manual** ☐ Disable

☐ Shared to other computers

Addresses

Address	Netmask	Gateway	
192.168.10.1	255.255.255.0		

DNS Automatic ☒

8.8.8.8

Separate IP addresses with commas

Router 설정

■ Ubuntu Network 2 설정

- Interface : ens35
- IPv4 Method : Manual
- IP : 192.168.128.254
- Netmask : 24
- Gateway : 192.168.128.2
- DNS : 8.8.8.8

The screenshot shows the 'Wired' network settings window in Ubuntu. The 'IPv4' tab is selected, showing the following configuration:

- IPv4 Method:** Manual (selected), Automatic (DHCP), Link-Local Only, Disable, Shared to other computers.
- Addresses:** A table with columns for Address, Netmask, and Gateway. The first row contains 192.168.128.254, 255.255.255.0, and 192.168.128.2.
- DNS:** A text field containing 8.8.8.8, with a toggle switch for 'Automatic' set to 'On'.

Address	Netmask	Gateway
192.168.128.254	255.255.255.0	192.168.128.2

Router 설정

■ Interface간 패킷 전달

➤ ens33 Interface와 ens35 Interface간 패킷 전송을 위한 Routing 기능 활성화

➤ \$ sudo vi /etc/sysctl.conf

```
# Uncomment the next two lines to enable Spoof protection (reverse-path filter)
# Turn on Source Address Verification in all interfaces to
# prevent some spoofing attacks
#net.ipv4.conf.default.rp_filter=1
#net.ipv4.conf.all.rp_filter=1

# Uncomment the next line to enable TCP/IP SYN cookies
# See http://lwn.net/Articles/277146/
# Note: This may impact IPv6 TCP sessions too
#net.ipv4.tcp_syncookies=1

# Uncomment the next line to enable packet forwarding for IPv4
net.ipv4.ip_forward=0

# Uncomment the next line to enable packet forwarding for IPv6
# Enabling this option disables Stateless Address Autoconfiguration
# based on Router Advertisements for this host
#net.ipv6.conf.all.forwarding=1
```

➤ #net.ipv4.ip_forwarding=1 행을 찾아서 맨 앞의 # 제거 후 저장

➤ 변경된 내용을 적용

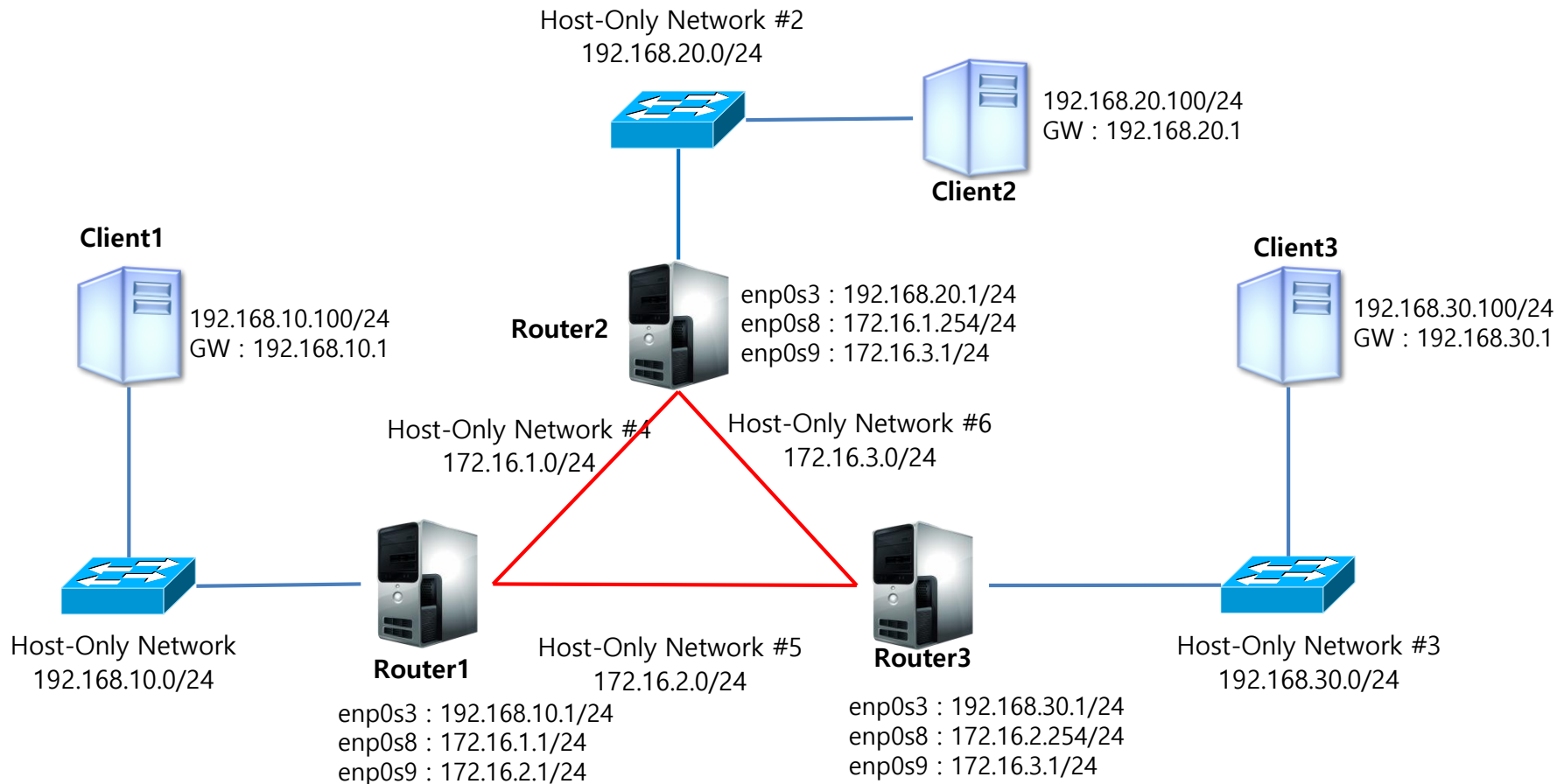
➤ \$ sudo sysctl -p /etc/sysctl.conf

```
user01@Server:~/Desktop$ sudo sysctl -p /etc/sysctl.conf
[sudo] password for user01:
net.ipv4.ip_forward = 1
user01@Server:~/Desktop$
```

2. RIP 라우팅 연결

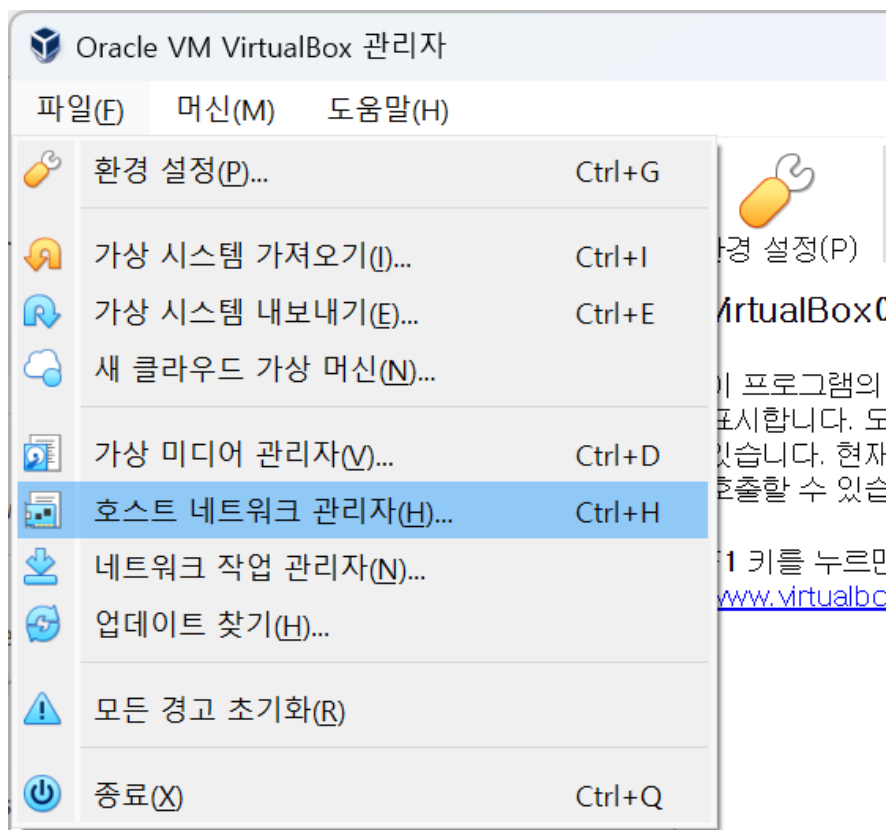
구성도

- 우분투 리눅스로 RIP라우팅 프로토콜을 실행하여 Dynamic Routing을 통해 상호 네트워크들간 통신을 할 수 있도록 설정하도록 합니다.



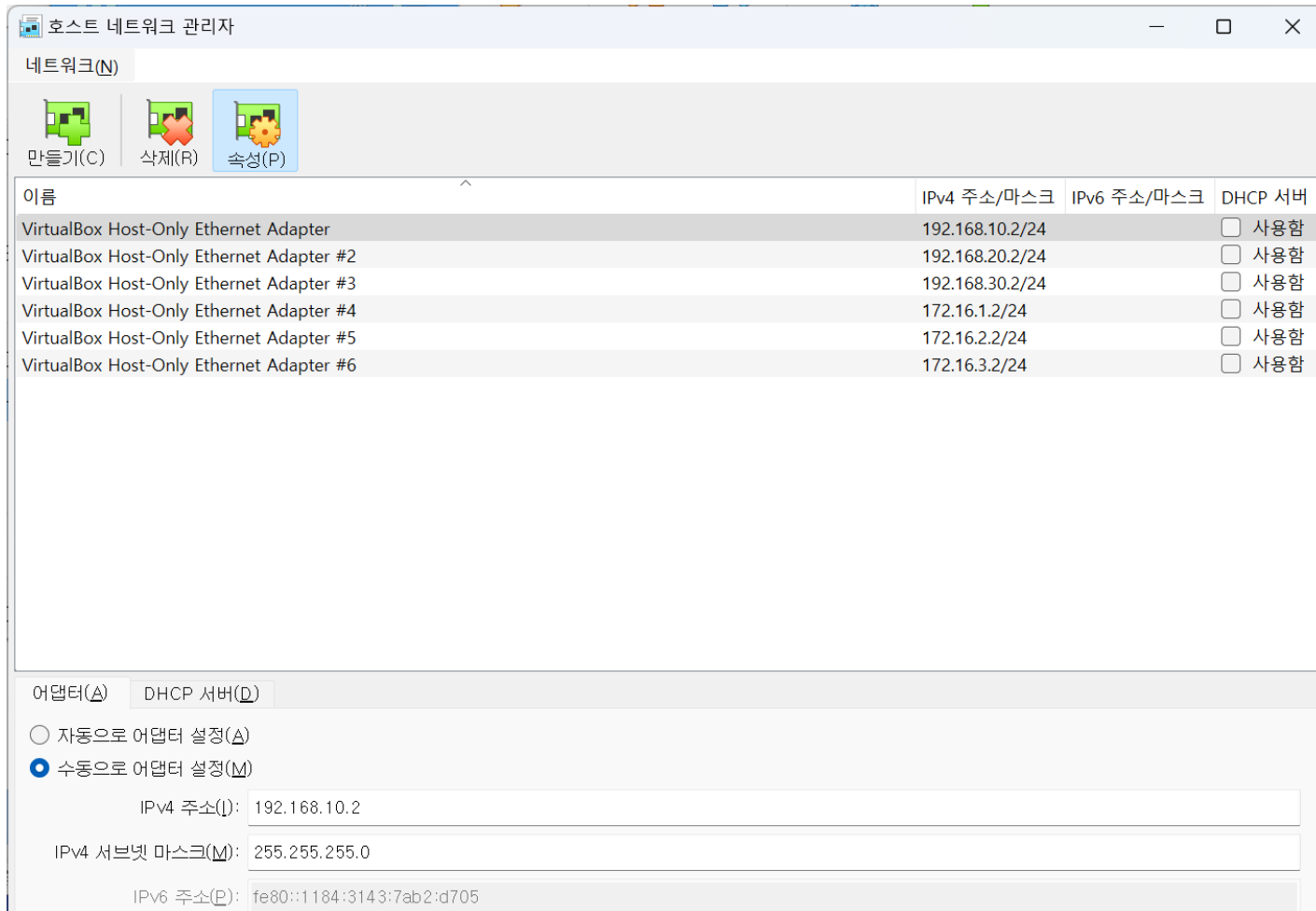
Network 환경 설정

- VirtualBox 프로그램에서 '파일(F)' → '호스트 네트워크 관리자(H)...'를 실행한다.



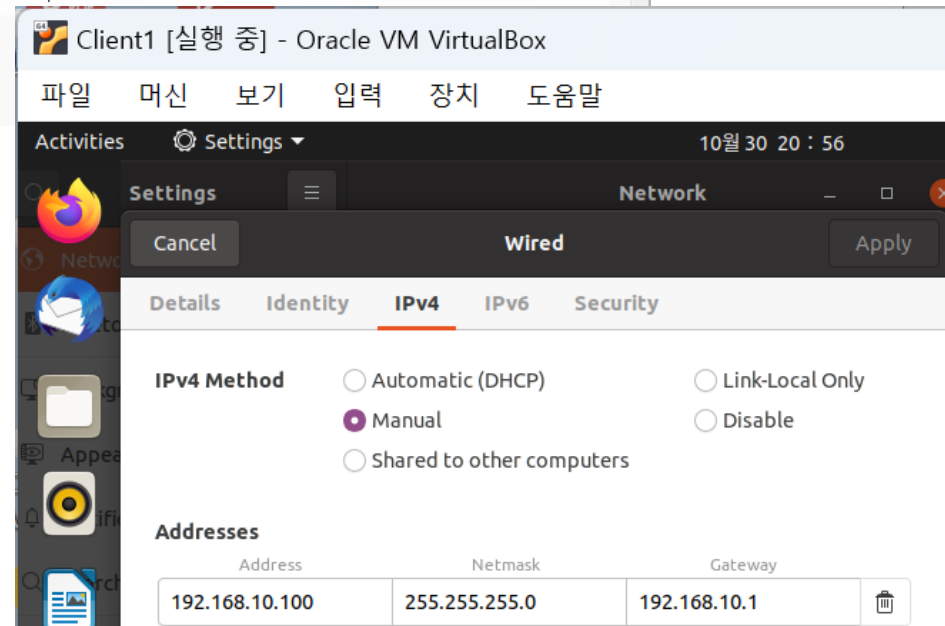
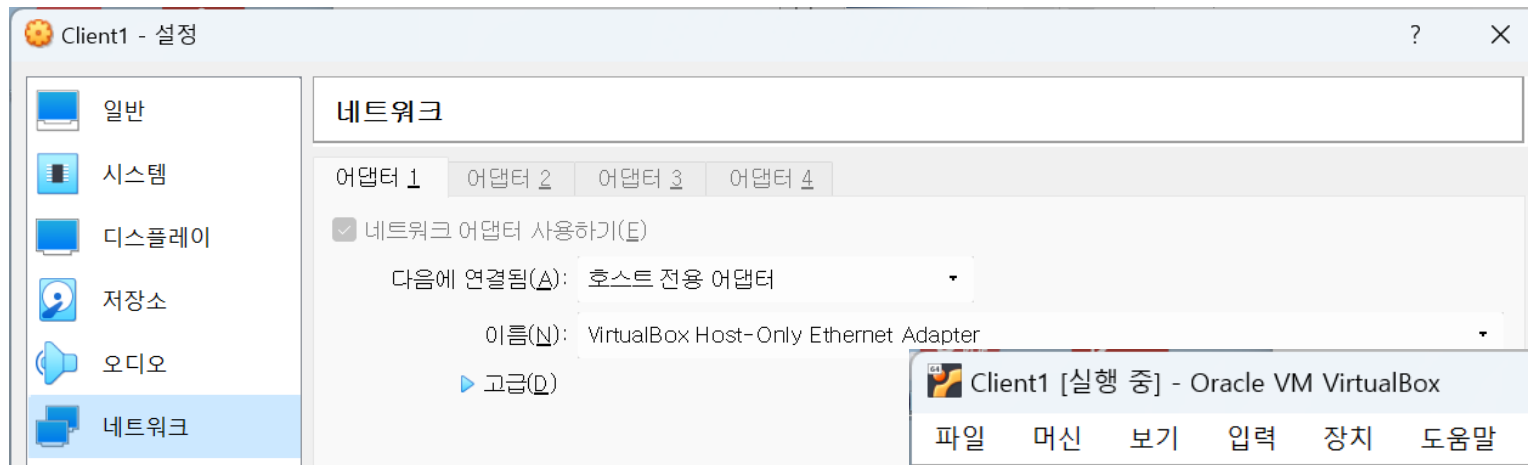
Network 환경 설정

- '만들기(C)'를 클릭하여 다음과 같이 VirtualBox Host-Only Ethernet Adapter를 #6번까지 생성하고 'IPv4 주소'와 '서브넷 마스크'를 수정해 준다.



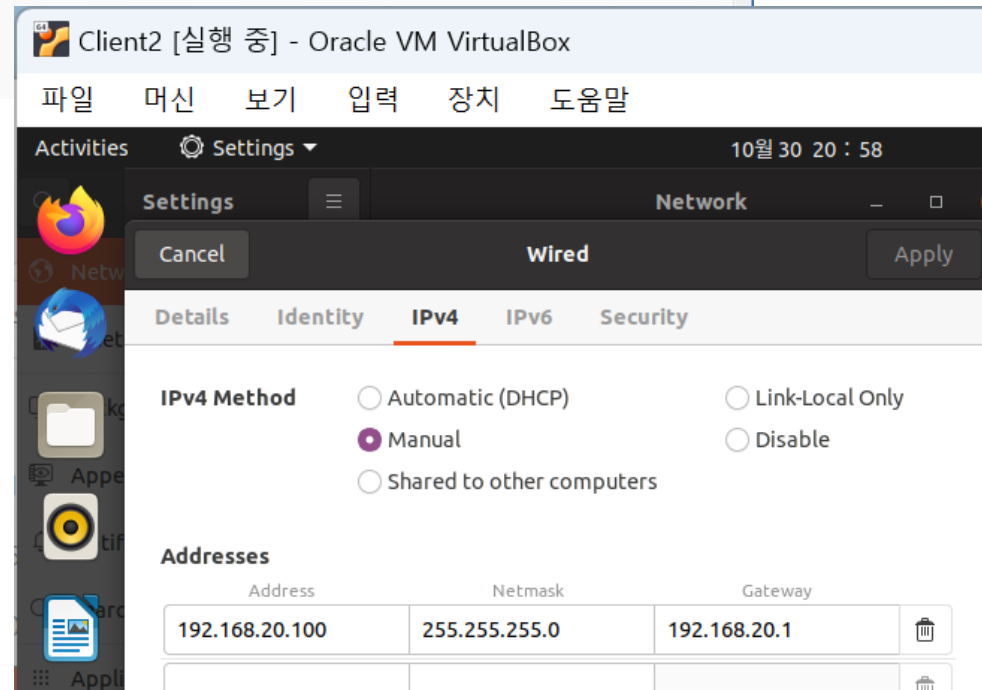
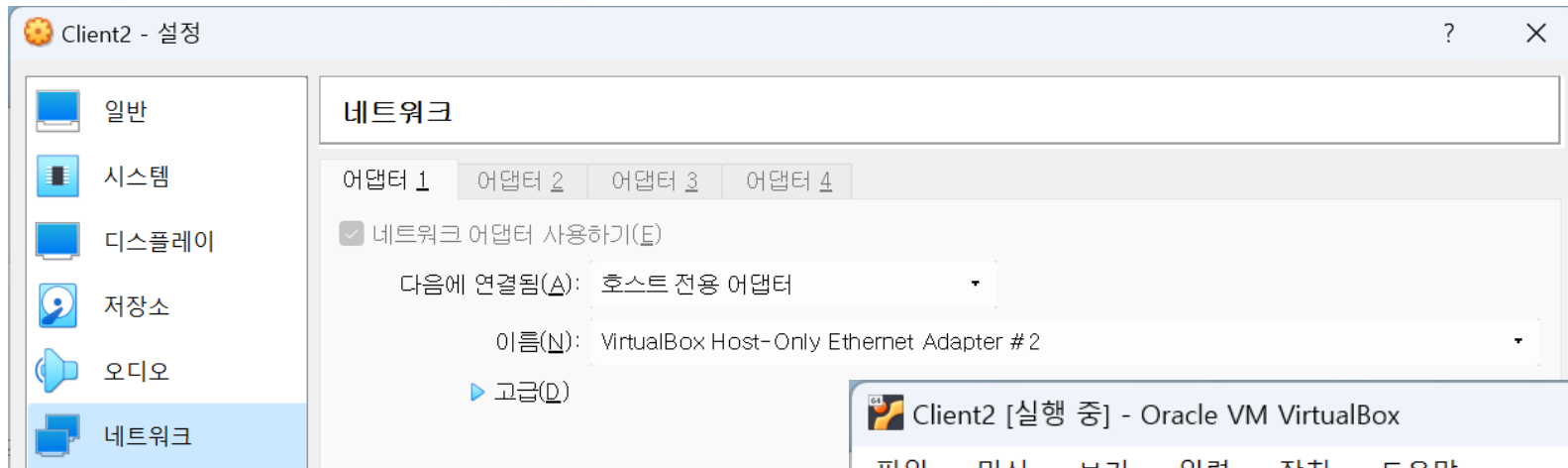
Client 설정

- Client1 이미지를 완전 복제로 설정하여 Client2, Client3 이미지를 생성한다.
- Client1의 정보를 다음과 같이 설정하도록 한다.



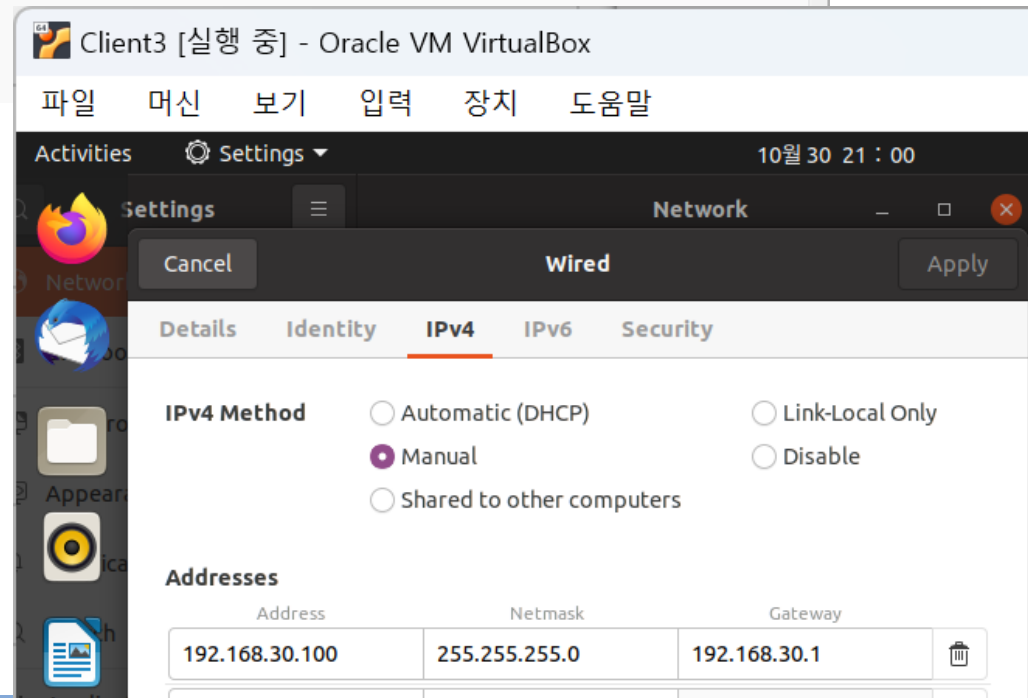
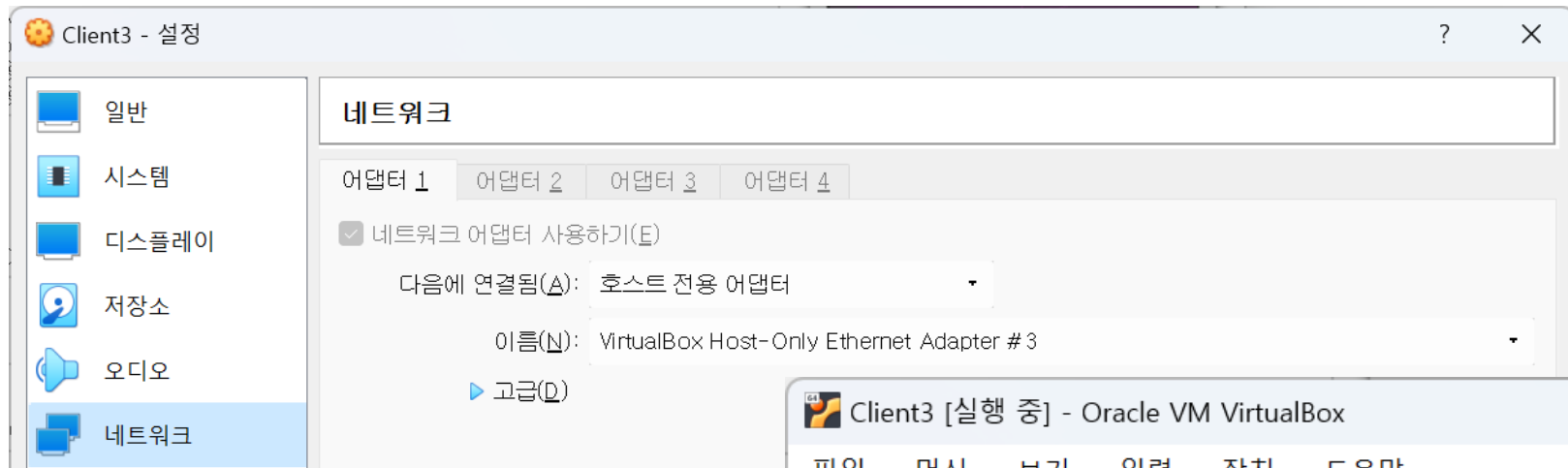
Client 설정

- Client2의 정보를 다음과 같이 설정하도록 한다.



Client 설정

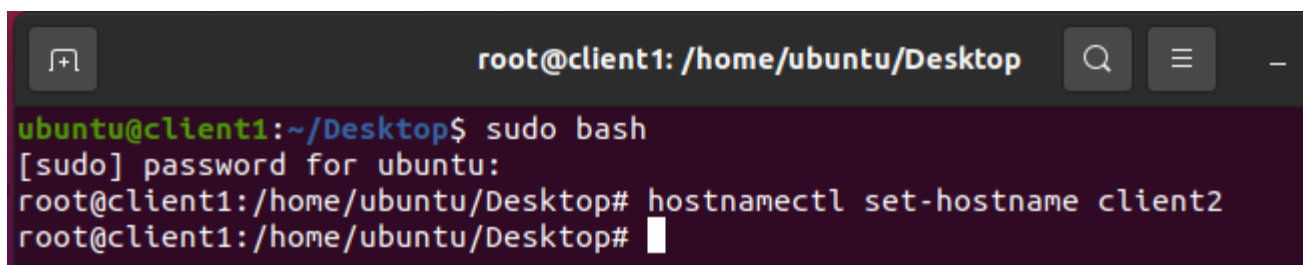
- Client3의 정보를 다음과 같이 설정하도록 한다.



Client 설정

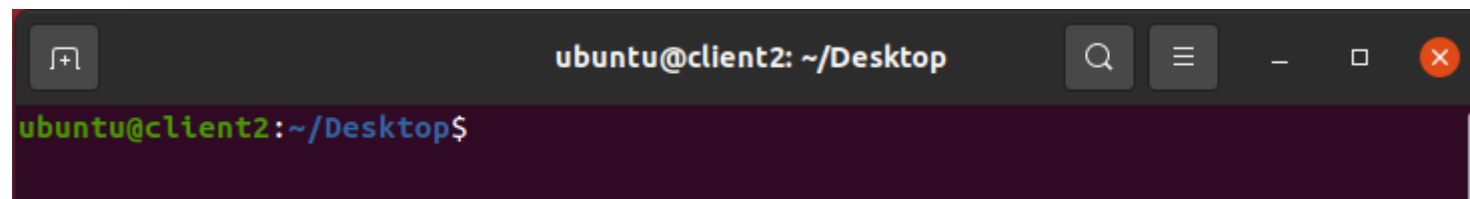
- Client를 실행하여 터미널창에서 Client의 이름을 변경해 준다.

- 관리자 권한으로 변경
- hostnamectl set-hostname 변경할이름



```
root@client1: /home/ubuntu/Desktop
ubuntu@client1:~/Desktop$ sudo bash
[sudo] password for ubuntu:
root@client1:/home/ubuntu/Desktop# hostnamectl set-hostname client2
root@client1:/home/ubuntu/Desktop#
```

- 터미널 창을 종료하고 다시 오픈하여 호스트 이름이 변경되었는지 확인한다.



```
ubuntu@client2: ~/Desktop
ubuntu@client2:~/Desktop$
```

Router 설정

- Router를 종료하고 다음과 같이 네트워크 카드를 2개 추가하고 호스트전용 어댑터에 배정하도록 한다.

 Client1 → 실행 중	일반 이름: Router1 운영 체제: Ubuntu (64-bit)
 Router1 ⏻ 전원 꺼짐	시스템 기본 메모리: 1024 MB 프로세서: 2 부팅 순서: 광 디스크, 하드 디스크 가속: VT-X/AMD-V, 네스티드 페이징, KVM 반가상화
 Client2 → 실행 중	디스플레이 비디오 메모리: 24 MB 그래픽 컨트롤러: VBoxSVGA 원격 데스크톱 서버: 사용 안 함 녹화: 사용 안 함
 Client3 → 실행 중	저장소 컨트롤러: IDE IDE 제컨더리 마스터: [광학 드라이브] 비어 있음 컨트롤러: SATA SATA 포트 0: Router1.vdi (일반, 20.00 GB)
 Router2 → 실행 중	오디오 사용 안 함
 Router3 → 실행 중	네트워크 어댑터 1: Intel PRO/1000 MT Desktop(82540EM) 어댑터 2: Intel PRO/1000 MT Desktop(82540EM) 어댑터 3: Intel PRO/1000 MT Desktop(82540EM)

Router 설정

- Router 이미지를 실행하고 quagga 패키지를 설치하도록 한다.
 - apt install quagga

```
root@router1: /home/ubuntu/Desktop
ubuntu@router1:~/Desktop$ sudo bash
root@router1:/home/ubuntu/Desktop# apt install quagga
Reading package lists... Done
Building dependency tree
Reading state information... Done
The following additional packages will be installed:
  quagga-bgpd quagga-core quagga-isisd quagga-ospf6d quagga-ospfd quagga-pimd
  quagga-ripd quagga-ripngd
Suggested packages:
  snmpd
The following NEW packages will be installed:
  quagga quagga-bgpd quagga-core quagga-isisd quagga-ospf6d quagga-ospfd
  quagga-pimd quagga-ripd quagga-ripngd
0 upgraded, 9 newly installed, 0 to remove and 0 not upgraded.
Need to get 1,276 kB of archives.
After this operation, 5,716 kB of additional disk space will be used.
Do you want to continue? [Y/n]
```

Router 설정

■ 기본 설정파일들을 생성해 준다.

➤ vi /etc/quagga/deamons

```
root@router1: /home/ubuntu/Desktop
zebra=1
ripd=1
~
```

➤ vi /etc/quagga/zebra.conf

```
root@router1: /home/ubuntu/Desktop
hostname router1
password quagga
enable password quagga
~
```

- hostname 은 동일하게
- password는 임의로

➤ vi /etc/quagga/ripd.conf

```
root@router1: /home/ubuntu/Desktop
hostname ripd
password quagga
~
```

Router 설정

- /etc/services 파일에 zebra와 ripd port 등록 후 재부팅

➤ vi /etc/ services

```
root@router1: /home/ubuntu/Desktop
rtcm-sc104      2101/tcp          # RTCM SC-104 IANA 1/29/99
rtcm-sc104      2101/udp
gsigatekeeper   2119/tcp
gris            2135/tcp          # Grid Resource Information Server
cvspserver      2401/tcp          # CVS client/server operations
venus           2430/tcp          # codacon port
venus           2430/udp          # Venus callback/wbc interface
venus-se        2431/tcp          # tcp side effects
venus-se        2431/udp          # udp sftp side effect
codasrv         2432/tcp          # not used
codasrv         2432/udp          # server port
codasrv-se      2433/tcp          # tcp side effects
codasrv-se      2433/udp          # udp sftp side effect
mon             2583/tcp          # MON traps
mon             2583/udp
zebra           2601/tcp
ripd            2602/tcp
dict            2628/tcp          # Dictionary server
f5-globalsite   2792/tcp
gsiftp          2811/tcp
gpsd            2947/tcp
gds-db          3050/tcp          gds_db      # InterBase server
icpv2           3130/tcp          icp          # Internet Cache Protocol
icpv2           3130/udp          icp
isns            3205/tcp          # iSNS Server Port
isns            3205/udp          # iSNS Server Port
iscsi-target    3260/tcp
```


Router 설정

■ quagga 접속 및 네트워크 정보 설정

```
ubuntu@router1: ~/Desktop

ubuntu@router1:~/Desktop$ telnet localhost zebra
Trying 127.0.0.1...
Connected to localhost.
Escape character is '^]'.

Hello, this is Quagga (version 1.2.4).
Copyright 1996-2005 Kunihiro Ishiguro, et al.

User Access Verification

Password:
router1> en
Password:
router1# conf t
router1(config)# int enp0s3
router1(config-if)# ip address 192.168.10.1/24
router1(config-if)# exit
router1(config)# int enp0s8
router1(config-if)# ip address 172.16.1.1/24
router1(config-if)# exit
router1(config)# int enp0s9
router1(config-if)# ip address 172.16.2.1/24
router1(config-if)# end
router1# write memory
Configuration saved to /etc/quagga/zebra.conf
router1#
```

Router 설정

■ 네트워크 정보 설정 확인

```
ubuntu@router1: ~/
router1# show running-config

Current configuration:
!
hostname router1
password quagga
enable password quagga
!
interface enp0s3
 ip address 192.168.10.1/24
!
interface enp0s8
 ip address 172.16.1.1/24
!
interface enp0s9
 ip address 172.16.2.1/24
!
interface lo
!
ip forwarding
!
!
line vty
!
end
router1# quit
Connection closed by foreign host.
ubuntu@router1:~/Desktop$
```

ubuntu@router1: ~/

```
ubuntu@router1: ~/Desktop
ubuntu@router1:~/Desktop$ sudo cat /etc/quagga/zebra.conf
[sudo] password for ubuntu:
!
! Zebra configuration saved from vty
! 2022/10/30 21:45:47
!
hostname router1
password quagga
enable password quagga
!
interface enp0s3
 ip address 192.168.10.1/24
!
interface enp0s8
 ip address 172.16.1.1/24
!
interface enp0s9
 ip address 172.16.2.1/24
!
interface lo
!
ip forwarding
!
!
line vty
!
end
ubuntu@router1:~/Desktop$
```

Router 설정

■ Rip 프로토콜 정보 설정

```
ubuntu@router1: ~/Desktop

ubuntu@router1:~/Desktop$ telnet localhost ripd
Trying 127.0.0.1...
Connected to localhost.
Escape character is '^]'.

Hello, this is Quagga (version 1.2.4).
Copyright 1996-2005 Kunihiro Ishiguro, et al.

User Access Verification

Password:
ripd> en
ripd# conf t
ripd(config)# router rip
ripd(config-router)# network 192.168.10.0/24
ripd(config-router)# network 172.16.1.0/24
ripd(config-router)# network 172.16.2.0/24
ripd(config-router)# end
ripd# write memory
Configuration saved to /etc/quagga/ripd.conf
ripd#
```

Router 설정

■ Rip 프로토콜 정보 확인

```
ripd#  
ripd# show running-config  
  
Current configuration:  
!  
hostname ripd  
password quagga  
!  
router rip  
 network 172.16.1.0/24  
 network 172.16.2.0/24  
 network 192.168.10.0/24  
!  
line vty  
!  
end  
ripd# quit  
Connection closed by foreign host.  
ubuntu@router1:~/Desktop$
```

```
ubuntu@router1: ~/Desktop  
ubuntu@router1:~/Desktop$ sudo cat /etc/quagga/ripd.conf  
!  
! Zebra configuration saved from vty  
!   2022/10/30 21:51:41  
!  
hostname ripd  
password quagga  
!  
router rip  
 network 172.16.1.0/24  
 network 172.16.2.0/24  
 network 192.168.10.0/24  
!  
line vty  
!  
ubuntu@router1:~/Desktop$
```

Router 설정

- OS 종료 후 Router2, Router3 이미지 복제 및 hostname 변경

?

×

← 가상 머신 선택

새 머신의 이름과 경로

새 가상 머신의 이름 및 폴더(선택 사항)를 입력하십시오. 새 머신은 머신 **Router1**의 복제본이 될 것입니다.

이름: Router2

경로: T:\WVMBBox

MAC 주소 정책(P): 모든 네트워크 어댑터의 새 MAC 주소 생성

추가 옵션:

모든 네트워크 어댑터 MAC 주소 포함
NAT 네트워크 어댑터 MAC 주소만 포함
모든 네트워크 어댑터의 새 MAC 주소 생성

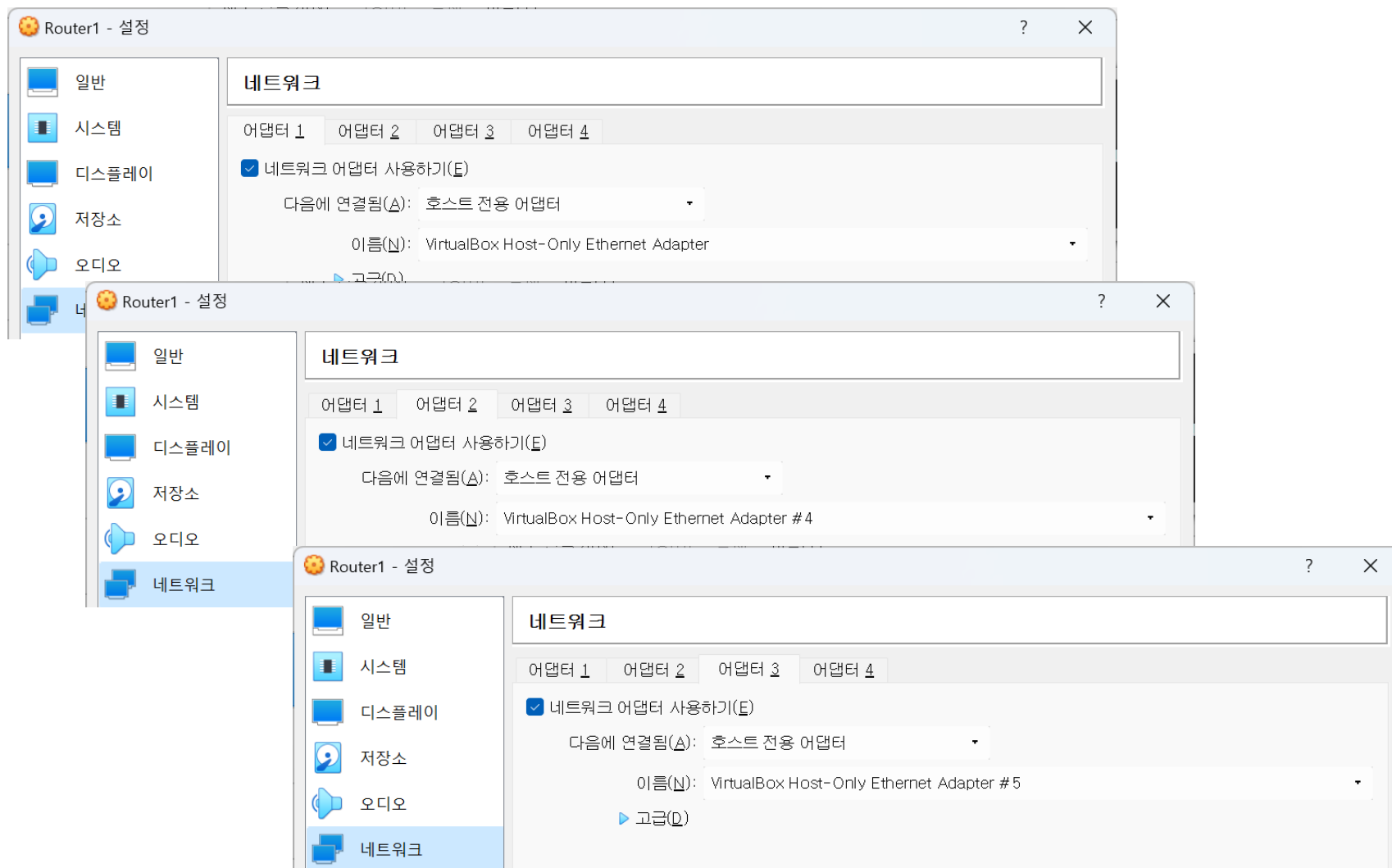
전문가 모드(E)

다음(N)

취소

Router1 설정

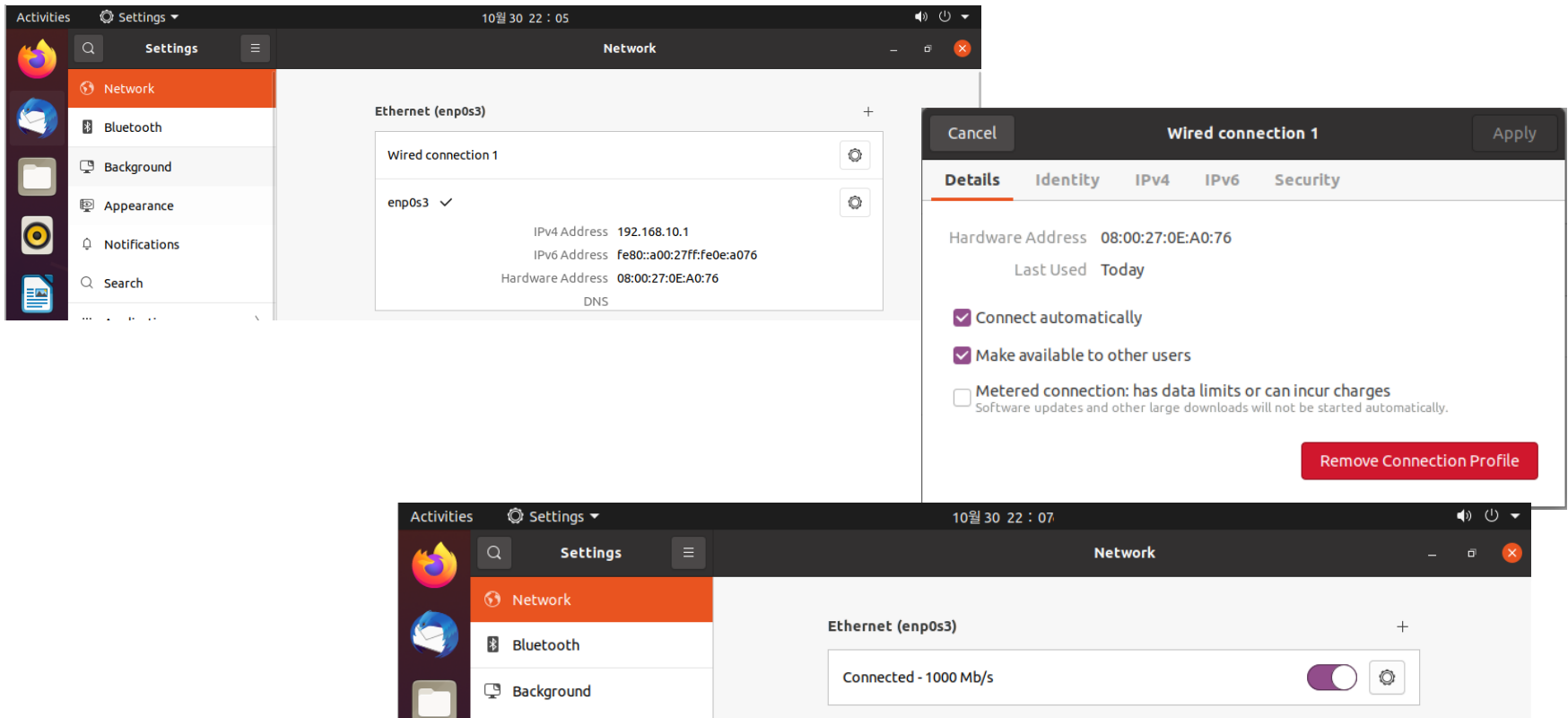
■ Router1 어댑터 설정



Router1 설정

■ Router1 IP 설정

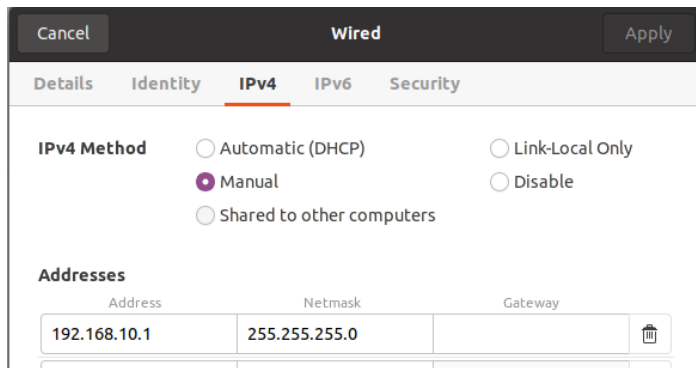
- 네트워크 연결 아이콘이 나타나지 않는 경우 다음과 같이 동일한 어댑터에 여러 프로파일이 적용되어 있을 수 있음
- '✓' 기호가 없는 프로파일을 선택하여 'Remove' 시키면 된다.



Router1 설정

■ Router1 IP 설정

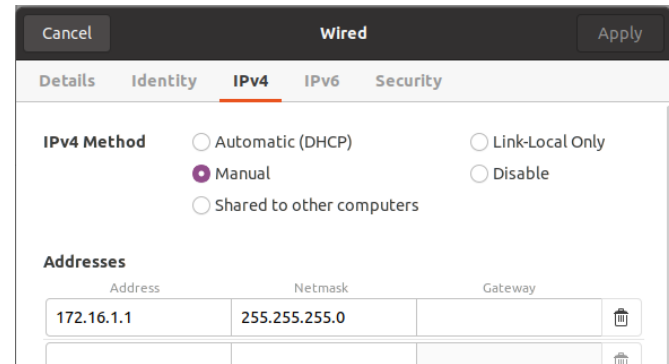
- 각각의 어댑터에 대해 다음과 같이 IP정보를 설정해 준다.



The screenshot shows the 'Wired' network settings window for the 'enp0s3' interface. The 'IPv4' tab is selected. Under 'IPv4 Method', the 'Manual' option is chosen. The 'Addresses' table contains one entry with the address '192.168.10.1' and netmask '255.255.255.0'. The 'Gateway' field is empty.

Address	Netmask	Gateway
192.168.10.1	255.255.255.0	

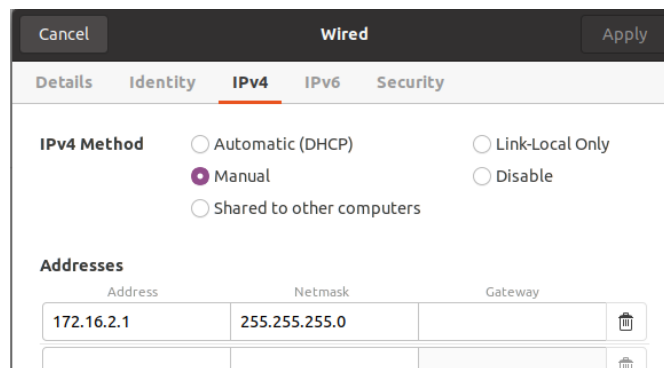
enp0s3



The screenshot shows the 'Wired' network settings window for the 'enp0s8' interface. The 'IPv4' tab is selected. Under 'IPv4 Method', the 'Manual' option is chosen. The 'Addresses' table contains one entry with the address '172.16.1.1' and netmask '255.255.255.0'. The 'Gateway' field is empty.

Address	Netmask	Gateway
172.16.1.1	255.255.255.0	

enp0s8



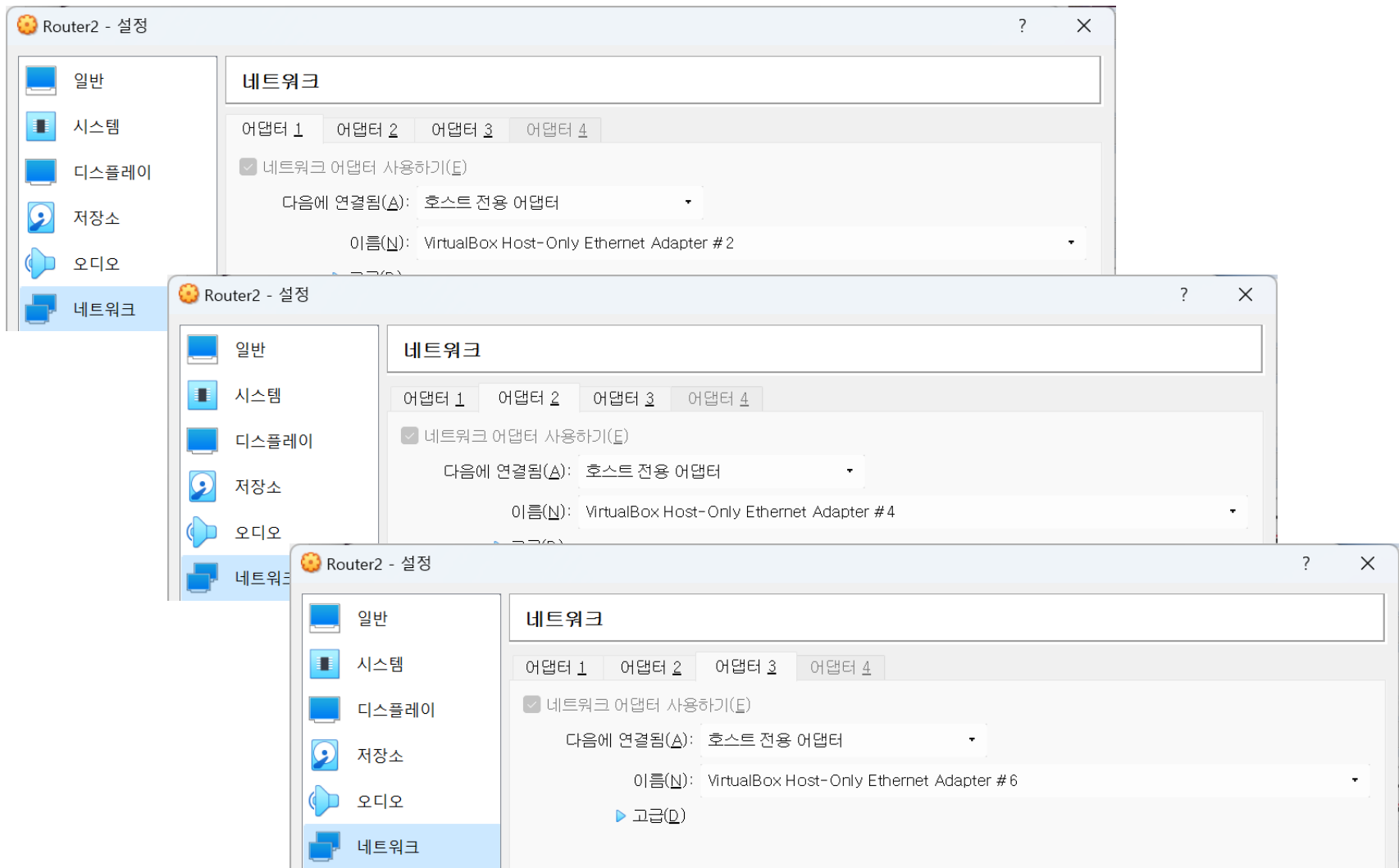
The screenshot shows the 'Wired' network settings window for the 'enp0s9' interface. The 'IPv4' tab is selected. Under 'IPv4 Method', the 'Manual' option is chosen. The 'Addresses' table contains one entry with the address '172.16.2.1' and netmask '255.255.255.0'. The 'Gateway' field is empty.

Address	Netmask	Gateway
172.16.2.1	255.255.255.0	

enp0s9

Router2 설정

■ Router2 어댑터 설정



Router2 설정

■ Router2 IP 설정

➤ 각각의 어댑터에 대해 다음과 같이 IP정보를 설정해 준다.

Wired

Cancel Apply

Details Identity **IPv4** IPv6 Security

IPv4 Method

☐ Automatic (DHCP) ☐ Link-Local Only

☒ Manual ☐ Disable

☐ Shared to other computers

Addresses

Address	Netmask	Gateway
192.168.20.1	255.255.255.0	

enp0s3

Wired

Cancel Apply

Details Identity **IPv4** IPv6 Security

IPv4 Method

☐ Automatic (DHCP) ☐ Link-Local Only

☒ Manual ☐ Disable

☐ Shared to other computers

Addresses

Address	Netmask	Gateway
172.16.1.254	255.255.255.0	

enp0s8

Wired

Cancel Apply

Details Identity **IPv4** IPv6 Security

IPv4 Method

☐ Automatic (DHCP) ☐ Link-Local Only

☒ Manual ☐ Disable

☐ Shared to other computers

Addresses

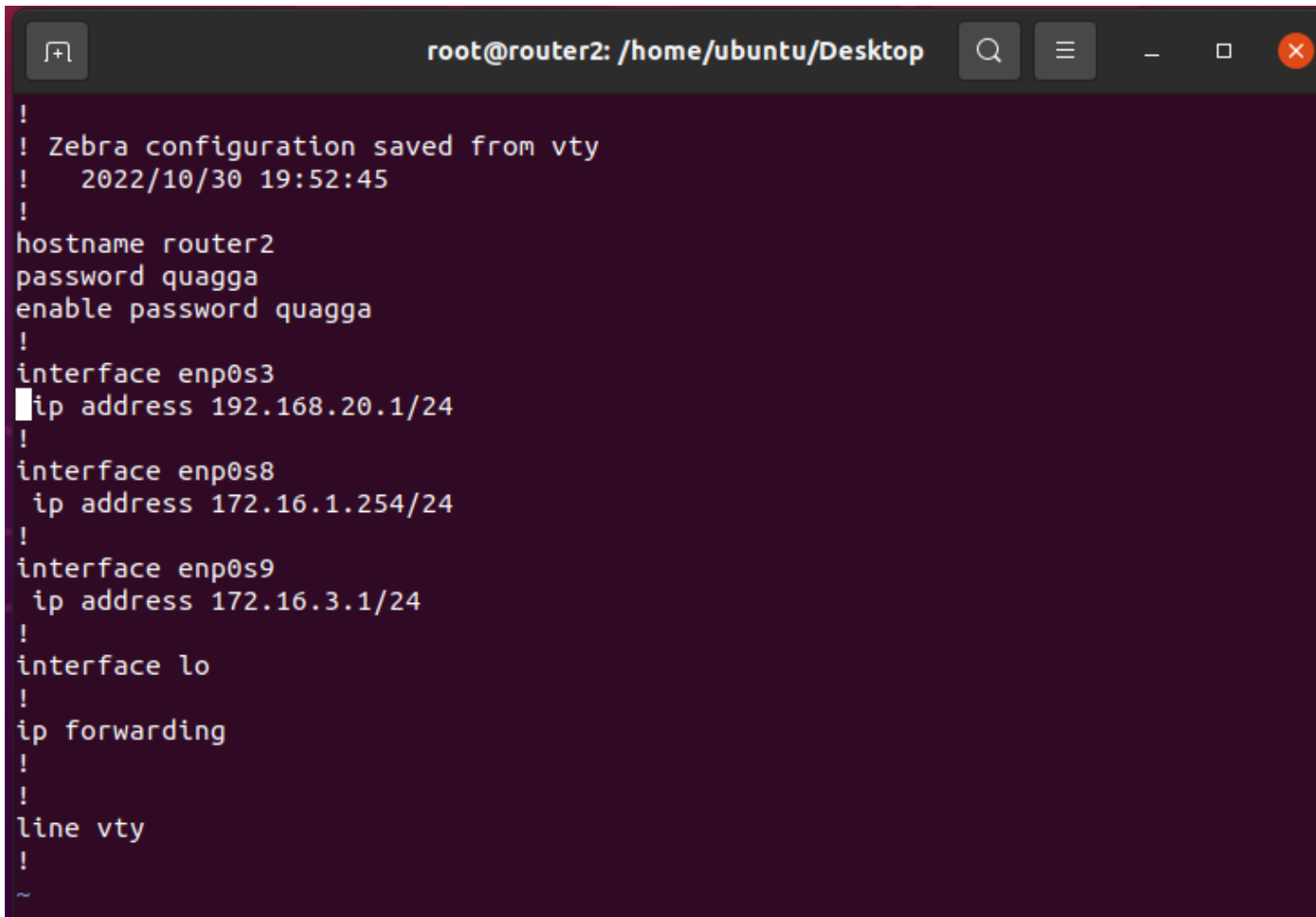
Address	Netmask	Gateway
172.16.3.1	255.255.255.0	

enp0s9

Router2 설정

■ Router2 quagga 라우터 네트워크 정보 수정

➤ vi /etc/quagga/zebra.conf

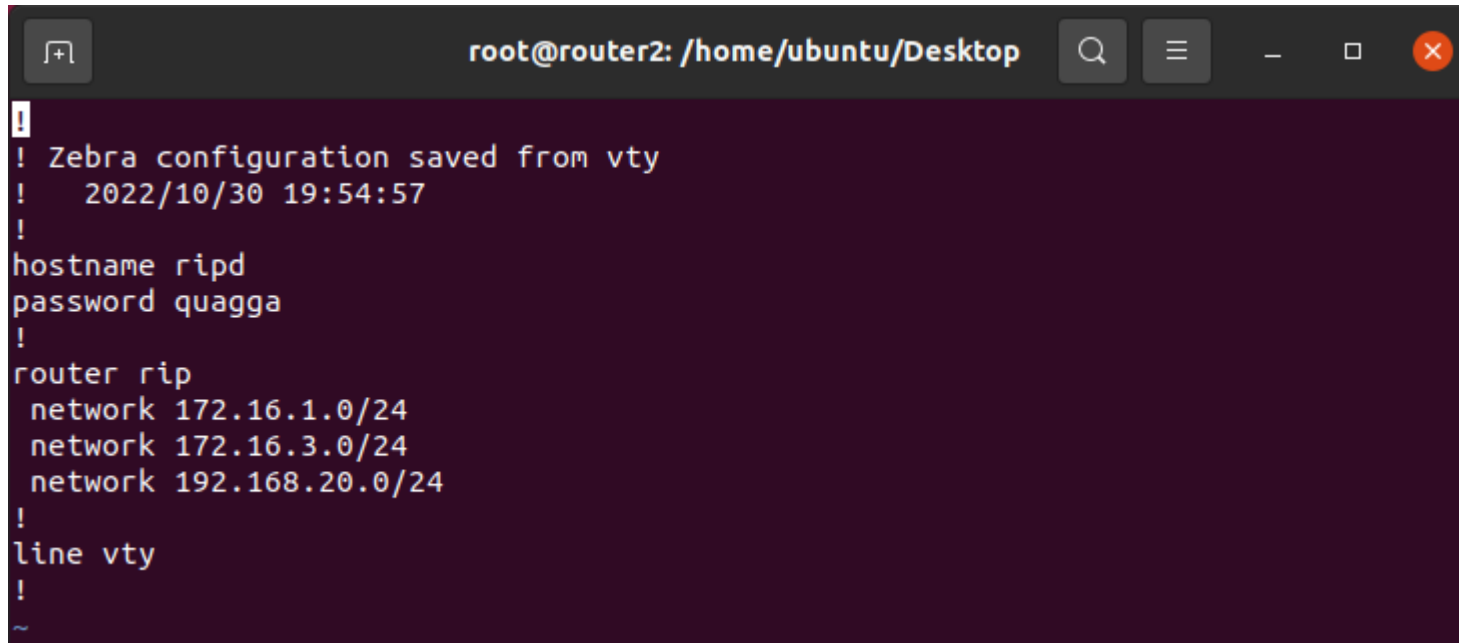


```
!
! Zebra configuration saved from vty
!   2022/10/30 19:52:45
!
hostname router2
password quagga
enable password quagga
!
interface enp0s3
ip address 192.168.20.1/24
!
interface enp0s8
ip address 172.16.1.254/24
!
interface enp0s9
ip address 172.16.3.1/24
!
interface lo
!
ip forwarding
!
!
line vty
!
~
```

Router2 설정

■ Router2 Rip 프로토콜 정보 수정

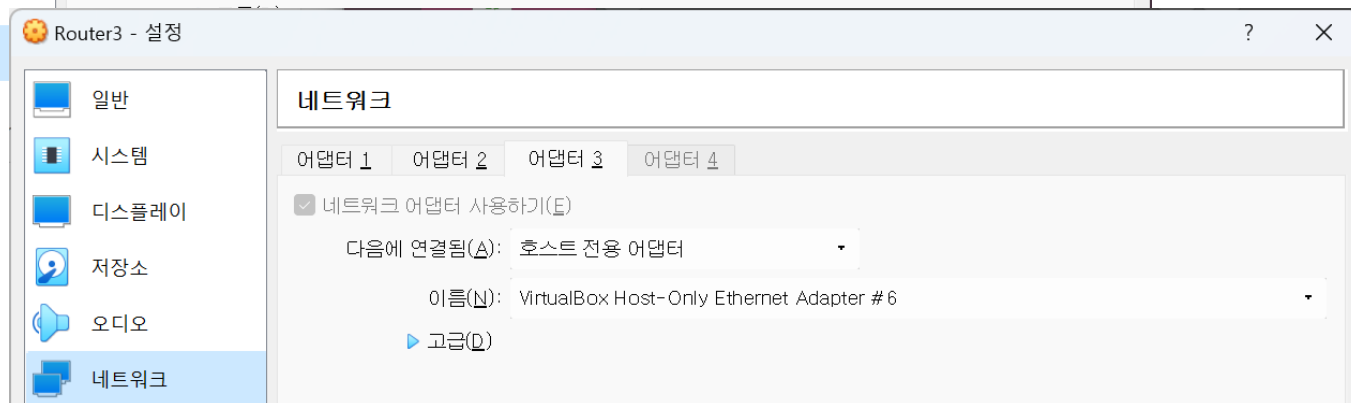
➤ vi /etc/quagga/ripd.conf



```
root@router2: /home/ubuntu/Desktop
!
! Zebra configuration saved from vty
!   2022/10/30 19:54:57
!
hostname ripd
password quagga
!
router rip
 network 172.16.1.0/24
 network 172.16.3.0/24
 network 192.168.20.0/24
!
line vty
!
~
```

Router3 설정

■ Router3 어댑터 설정



Router3 설정

■ Router3 IP 설정

➤ 각각의 어댑터에 대해 다음과 같이 IP정보를 설정해 준다.

Cancel Wired Apply

Details Identity **IPv4** IPv6 Security

IPv4 Method

☐ Automatic (DHCP) ☐ Link-Local Only

☒ Manual ☐ Disable

☐ Shared to other computers

Addresses

Address	Netmask	Gateway	
192.168.30.1	255.255.255.0		

enp0s3

Cancel Wired Apply

Details Identity **IPv4** IPv6 Security

IPv4 Method

☐ Automatic (DHCP) ☐ Link-Local Only

☒ Manual ☐ Disable

☐ Shared to other computers

Addresses

Address	Netmask	Gateway	
172.16.2.254	255.255.255.0		

enp0s8

Cancel Wired Apply

Details Identity **IPv4** IPv6 Security

IPv4 Method

☐ Automatic (DHCP) ☐ Link-Local Only

☒ Manual ☐ Disable

☐ Shared to other computers

Addresses

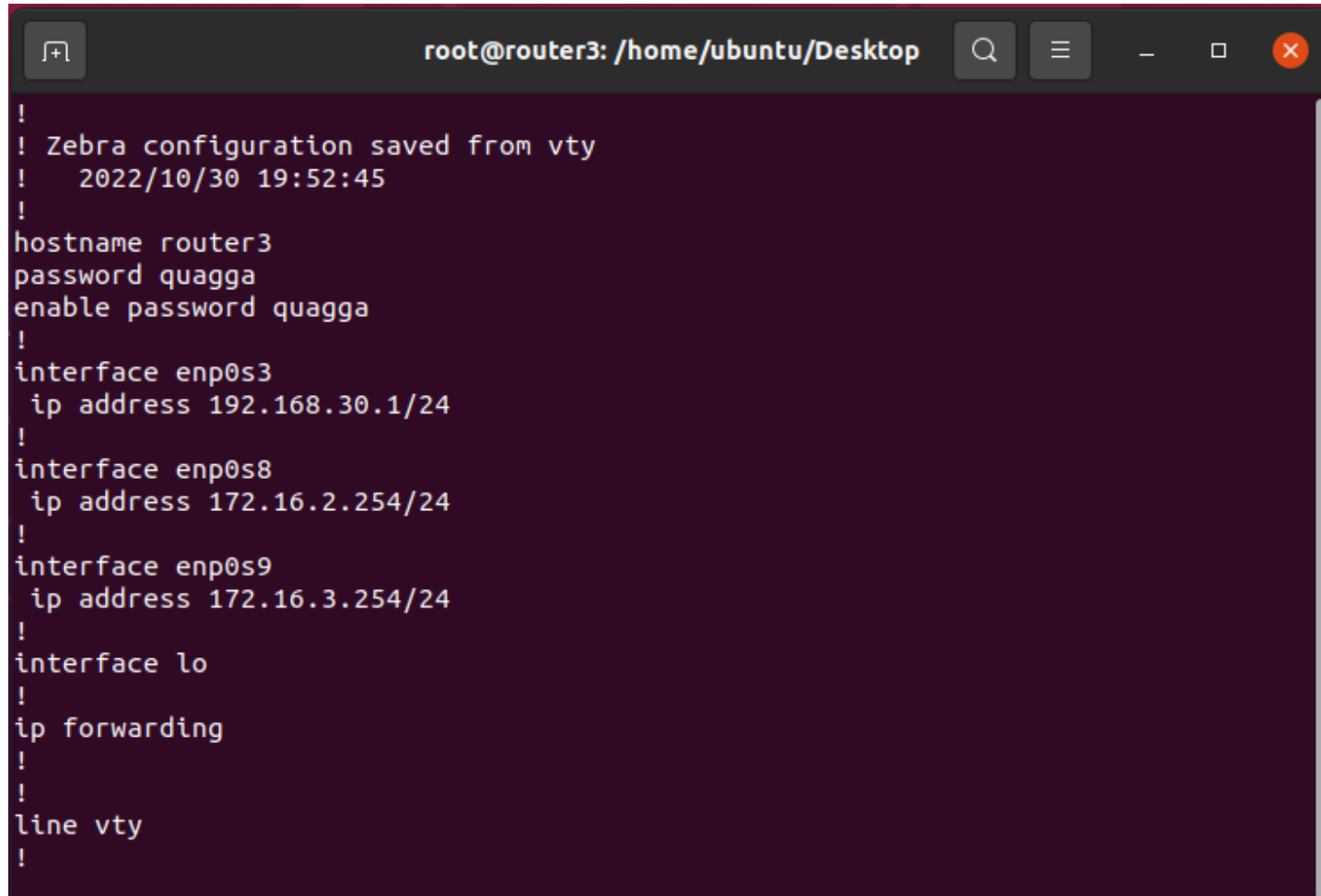
Address	Netmask	Gateway	
172.16.3.254	255.255.255.0		

enp0s9

Router3 설정

■ Router3 quagga 라우터 네트워크 정보 수정

➤ vi /etc/quagga/zebra.conf

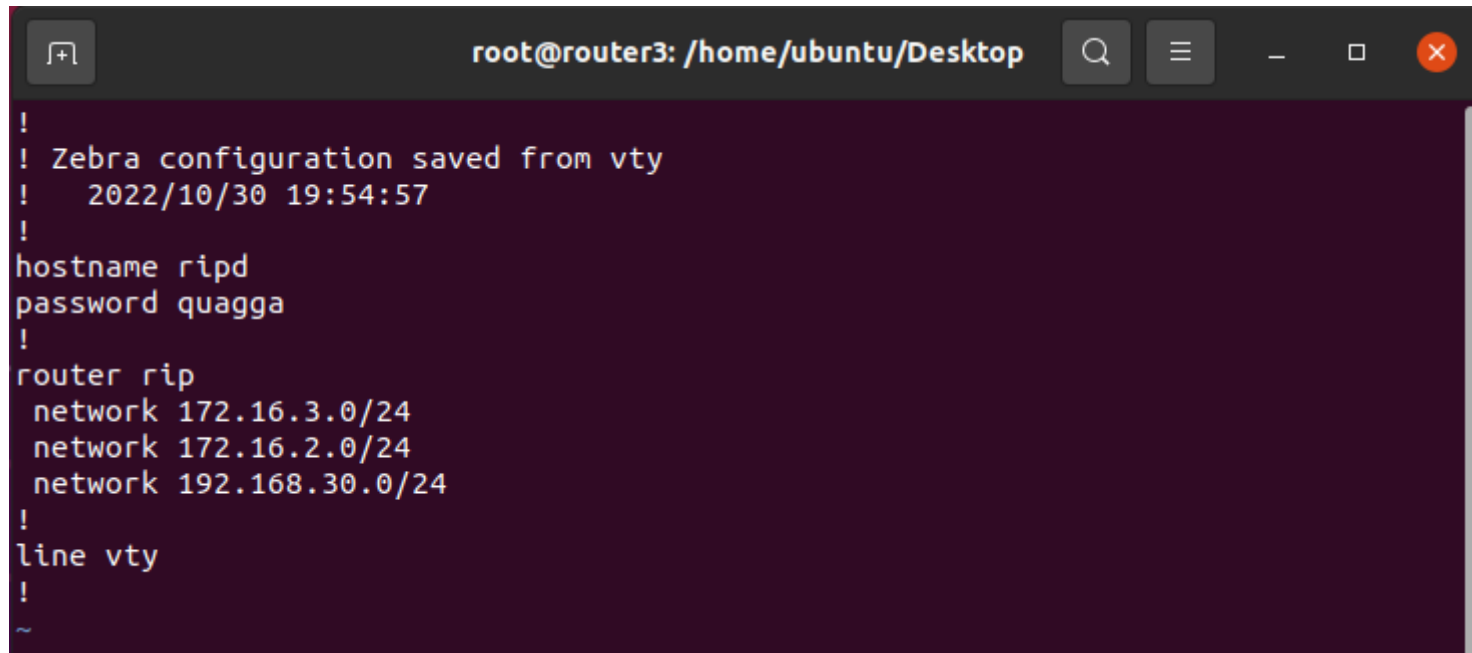


```
!
! Zebra configuration saved from vty
! 2022/10/30 19:52:45
!
hostname router3
password quagga
enable password quagga
!
interface enp0s3
 ip address 192.168.30.1/24
!
interface enp0s8
 ip address 172.16.2.254/24
!
interface enp0s9
 ip address 172.16.3.254/24
!
interface lo
!
ip forwarding
!
!
line vty
!
```

Router3 설정

■ Router3 Rip 프로토콜 정보 수정

➤ vi /etc/quagga/ripd.conf



```
root@router3: /home/ubuntu/Desktop
!
! Zebra configuration saved from vty
!   2022/10/30 19:54:57
!
hostname ripd
password quagga
!
router rip
 network 172.16.3.0/24
 network 172.16.2.0/24
 network 192.168.30.0/24
!
line vty
!
~
```


통신 테스트

- 모든 라우터를 재부팅 하도록 한다.
- Router3 에서 라우팅 정보를 확인

➤ ip route show

```
root@router3: /home/ubuntu/Desktop
root@router3:/home/ubuntu/Desktop# ip route show
169.254.0.0/16 dev enp0s3 scope link metric 1000
172.16.1.0/24 via 172.16.3.1 dev enp0s9 proto zebra metric 20
172.16.2.0/24 dev enp0s8 proto kernel scope link src 172.16.2.254
172.16.3.0/24 dev enp0s9 proto kernel scope link src 172.16.3.254
192.168.10.0/24 via 172.16.2.1 dev enp0s8 proto zebra metric 20
192.168.20.0/24 via 172.16.3.1 dev enp0s9 proto zebra metric 20
192.168.30.0/24 dev enp0s3 proto kernel scope link src 192.168.30.1
root@router3:/home/ubuntu/Desktop#
```

➤ netstat -rn

```
root@router3: /home/ubuntu/Desktop
root@router3:/home/ubuntu/Desktop# netstat -rn
Kernel IP routing table
Destination      Gateway          Genmask         Flags   MSS Window  irtt  Iface
169.254.0.0      0.0.0.0         255.255.0.0     U        0 0        0 enp0s3
172.16.1.0       172.16.3.1     255.255.255.0   UG        0 0        0 enp0s9
172.16.2.0       0.0.0.0         255.255.255.0   U        0 0        0 enp0s8
172.16.3.0       0.0.0.0         255.255.255.0   U        0 0        0 enp0s9
192.168.10.0     172.16.2.1     255.255.255.0   UG        0 0        0 enp0s8
192.168.20.0     172.16.3.1     255.255.255.0   UG        0 0        0 enp0s9
192.168.30.0     0.0.0.0         255.255.255.0   U        0 0        0 enp0s3
root@router3:/home/ubuntu/Desktop#
```

■ Client2 에서 ping test 수행

```
ubuntu@client2: ~/Desktop
ubuntu@client2:~/Desktop$ ping 192.168.10.100 -c 2
PING 192.168.10.100 (192.168.10.100) 56(84) bytes of data.
64 bytes from 192.168.10.100: icmp_seq=1 ttl=62 time=0.946 ms
64 bytes from 192.168.10.100: icmp_seq=2 ttl=62 time=0.960 ms

--- 192.168.10.100 ping statistics ---
2 packets transmitted, 2 received, 0% packet loss, time 1003ms
rtt min/avg/max/mdev = 0.946/0.953/0.960/0.007 ms
ubuntu@client2:~/Desktop$ ping 192.168.30.100 -c 2
PING 192.168.30.100 (192.168.30.100) 56(84) bytes of data.
64 bytes from 192.168.30.100: icmp_seq=1 ttl=62 time=0.721 ms
64 bytes from 192.168.30.100: icmp_seq=2 ttl=62 time=0.561 ms

--- 192.168.30.100 ping statistics ---
2 packets transmitted, 2 received, 0% packet loss, time 1003ms
rtt min/avg/max/mdev = 0.561/0.641/0.721/0.080 ms
ubuntu@client2:~/Desktop$ ping 172.16.2.254 -c 2
PING 172.16.2.254 (172.16.2.254) 56(84) bytes of data.
64 bytes from 172.16.2.254: icmp_seq=1 ttl=63 time=0.835 ms
64 bytes from 172.16.2.254: icmp_seq=2 ttl=63 time=1.15 ms

--- 172.16.2.254 ping statistics ---
2 packets transmitted, 2 received, 0% packet loss, time 1013ms
rtt min/avg/max/mdev = 0.835/0.990/1.146/0.155 ms
ubuntu@client2:~/Desktop$
```

End of Document